

Bayesian Logistic Regression for Loan Default Prediction on the Lending Club Dataset

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1 Introduction

Peer-to-peer lending lets individuals fund consumer loans directly. Whether those loans get paid back is what determines whether the platform survives. I fit a Bayesian logistic regression on Lending Club data to predict default, and then check that the inference is calibrated by running a simulation study.

A logistic regression is not the best model for this problem. A boosted tree would beat it on accuracy without much effort. The reason to use it anyway is that the coefficients have meaning and the posterior tells me how uncertain those meanings are. The simulation study is what makes that worth trusting. By generating data where I already know the true coefficients and seeing whether the posterior covers them at the rate it claims, I can rule out an implementation error that would not be visible from the real data fit alone.

2 Data

The dataset is Nathan George's Lending Club archive on Kaggle (<https://www.kaggle.com/datasets/wordsforthewise/lending-club>). It has every loan Lending Club originated from 2007 through Q4 2018, 2,260,701 rows in total with 151 columns. After keeping only loans with a final outcome (Fully Paid, Charged Off, or Default), parsing the percent-string columns, building a binary default indicator, and dropping rows with any missing values on the variables I needed, the cleaned set has 1,261,262 loans. The default rate is 19.5%.

I selected eight covariates that are observable when the loan is funded:

- `loan_amnt`: requested loan amount
- `int_rate`: interest rate Lending Club assigned
- `log_income`: log of annual income
- `dti`: debt-to-income ratio
- `fico`: average of the FICO range at origination
- `revol_util`: revolving credit utilization
- `emp_length`: years at current employer
- `installment`: required monthly payment

All eight are continuous. I standardized them to mean zero and unit variance. This puts the prior on a comparable scale across covariates and improves MCMC stability.

2.1 Exploratory observations

Table 1 reports summary statistics for the eight covariates after cleaning but before standardization.

Variable	Mean	SD	Min	Median	Max
<code>loan_amnt</code>	14,601.76	8,742.55	500.00	12,075.00	40,000.00
<code>int_rate</code>	13.22	4.77	5.31	12.74	30.99
<code>log_income</code>	11.11	0.52	6.40	11.08	16.21
<code>dti</code>	18.08	8.57	0.00	17.52	99.92
<code>fico</code>	698.17	31.68	627.00	692.00	847.50
<code>revol_util</code>	51.87	24.32	0.00	52.30	100.00
<code>emp_length</code>	5.97	3.69	0.00	6.00	10.00
<code>installment</code>	442.90	262.28	4.93	379.44	1,719.83

Table 1: Summary statistics for the eight covariates on the cleaned dataset ($n = 1,261,262$).

Two issues worth mentioning. First, the classes are imbalanced (80/20 split). The Bayesian fit is not affected because the likelihood isn't optimizing accuracy, but it shapes interpretation. Second, `installment` is mechanically determined by `loan_amnt`, `int_rate`, and the loan term. So `installment` is partially redundant once the first two are included, and that shows up later in the posterior in a way that requires interpretation.

3 Statistical Model

Each loan i has a binary outcome $y_i \in \{0, 1\}$ and a covariate vector $x_i \in \mathbb{R}^p$ with $p = 9$ (eight covariates plus an intercept). The model is

$$y_i \mid x_i, \beta \sim \text{Bernoulli}(\pi_i), \quad (1)$$

$$\pi_i = \frac{1}{1 + \exp(-x_i^\top \beta)}. \quad (2)$$

The likelihood for n independent observations is

$$p(y \mid X, \beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}. \quad (3)$$

For the prior I use a multivariate normal on the coefficient vector, centered at zero with diagonal covariance:

$$\beta \sim \mathcal{N}_p(0, \tau^2 I_p), \quad \tau^2 = 100. \quad (4)$$

This is weakly informative. On standardized covariates, a coefficient of magnitude 1 already implies an odds ratio close to 2.7 for a one-standard-deviation change. With $\tau = 10$ the prior allows coefficients up to about ± 30 before it really pushes back, which is far beyond anything physically reasonable. The goal is to keep the posterior proper without imposing strong prior beliefs.

The posterior (up to a constant) is

$$p(\beta \mid y, X) \propto \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \exp\left(-\frac{\beta^\top \beta}{2\tau^2}\right). \quad (5)$$

This has no closed form. A logistic likelihood and a normal prior do not produce a recognizable conjugate posterior, so I sample with Metropolis-Hastings.

4 Synthetic Data Generation

The simulation study needs synthetic datasets where I know the true β . The procedure:

1. Pick true coefficient values β^* . I used the maximum likelihood estimates from a frequentist

`glm()` fit on the full cleaned data. This grounds the simulation in a realistic parameter regime rather than arbitrary values.

2. Draw $n = 10,000$ rows of synthetic covariates. Each continuous variable comes from a marginal distribution chosen to roughly match the real data: gamma for the right-skewed ones (`loan_amnt`, `installment`), normal for the others. Bounded variables (`revol_util`, `emp_length`) get clipped.
3. Standardize the design matrix, prepend an intercept column.
4. Compute π_i from β^* and the synthetic X , then draw $y_i \sim \text{Bernoulli}(\pi_i)$.

The intercept of $\beta_0^* = -1.575$ produces synthetic datasets with a default rate around 18%, similar to the 19.5% in the real data.

5 MCMC Algorithm

I sample from the posterior with random-walk Metropolis-Hastings. At iteration t , given $\beta^{(t)}$, propose

$$\beta^* \sim \mathcal{N}_p\left(\beta^{(t)}, \Sigma_{\text{prop}}\right), \quad (6)$$

and accept with probability

$$\alpha = \min\left(1, \exp\left[\log p(\beta^* \mid y, X) - \log p(\beta^{(t)} \mid y, X)\right]\right). \quad (7)$$

Working on the log scale is necessary because the unnormalized posterior values can be too small to represent at this sample size. The proposal is symmetric so the Hastings ratio is one.

The proposal covariance Σ_{prop} is updated periodically from the empirical covariance of recent samples, scaled by $2.4^2/p$ following Roberts and Rosenthal’s optimal scaling for high-dimensional random walks. A fixed isotropic proposal performs poorly because `loan_amnt`, `int_rate`, and `installment` are correlated in the posterior. Adapting during burn-in allows the proposal to reflect the correlation structure of the posterior.

5.1 Implementing without R packages

The rubric prohibits packages. Since multivariate normal sampling and density evaluation are not in base R, I wrote both from scratch using the Cholesky factorization. Sampling $\mathcal{N}(\mu, \Sigma)$: draw $z \sim \mathcal{N}(0, I)$, compute the Cholesky factor L of Σ , return $\mu + Lz$. Evaluating the log-density at x :

$$-\frac{1}{2} \left(p \log(2\pi) + \log |\Sigma| + (x - \mu)^\top \Sigma^{-1} (x - \mu) \right), \quad (8)$$

where $\log |\Sigma| = 2 \sum_i \log L_{ii}$ and the quadratic form is computed with a triangular solve so I never have to form Σ^{-1} . Both functions live in `helpers.r`.

6 Simulation Study

The point of the simulation study is to verify that the posterior is calibrated. If it is, then over many synthetic datasets, the posterior mean is approximately unbiased and 95% credible intervals contain the true parameter about 95% of the time. Failure of calibration would indicate either a coding error or a problem with the sampler.

6.1 Design

I generated $N = 200$ synthetic datasets. Each uses the same true β^* but a different seed for the covariates and outcomes. Each dataset has $n = 10,000$ rows.

The choice of $N = 200$ comes from the Monte Carlo error on the coverage estimate. With $\hat{c} \approx 0.95$, the standard error is $\sqrt{\hat{c}(1 - \hat{c})/N} \approx 0.0154$ at $N = 200$. That's tight enough to distinguish true coverage of 0.95 from 0.90 but small enough to run on the cluster in minutes. Increasing to $N = 500$ would tighten the coverage estimate further but at substantial additional compute cost.

The choice of $n = 10,000$ matches the size of the subsample I use for the real data fit. The full cleaned dataset has 1.26 million rows, but running 200 MH chains at that scale is computationally prohibitive. Subsampling to 10,000 keeps each replicate's runtime under a minute and the posterior is still concentrated (posterior standard deviations all under 0.1).

Each MH chain ran for 20,000 iterations with a 5,000-iteration burn-in, leaving 15,000 post-burn-in samples per parameter. The simulation ran as a 200-task LSF array job on the NC State Hazel cluster.

6.2 Results

Across all 200 replicates, the acceptance rate stayed in $[0.277, 0.300]$ with mean 0.286 and standard deviation 0.004. The optimal rate for a 9-dimensional random walk is around 0.234, so the chains are mixing well; the proposal is slightly more conservative than necessary.

Figure 1 shows the sampling distribution of the posterior mean across replicates for each parameter, with red lines at the true values. The distributions are tight and well-centered. Bias is below 0.005 for every parameter (Table 2).

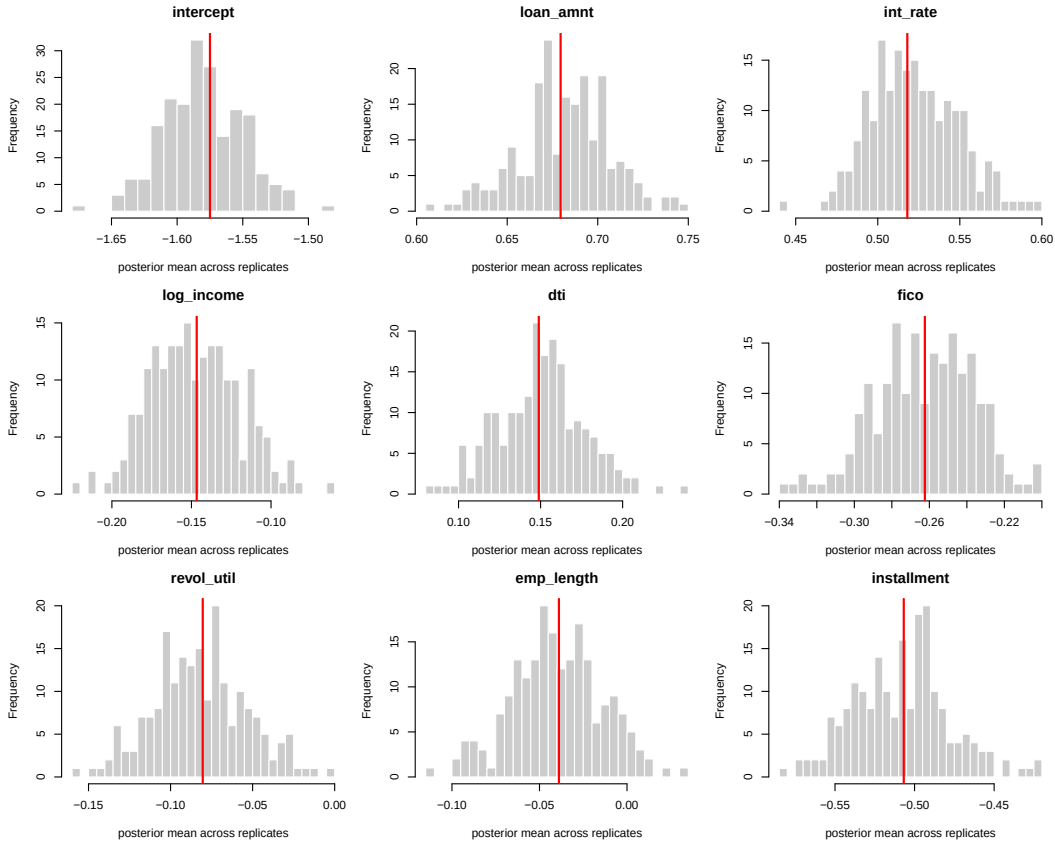


Figure 1: Sampling distribution of the posterior mean across $N = 200$ replicates. Red lines mark the true parameter values β^* .

Figure 2 shows empirical coverage of the 95% credible intervals across the 200 replicates. Every

parameter is within Monte Carlo error of the nominal 0.95 line.

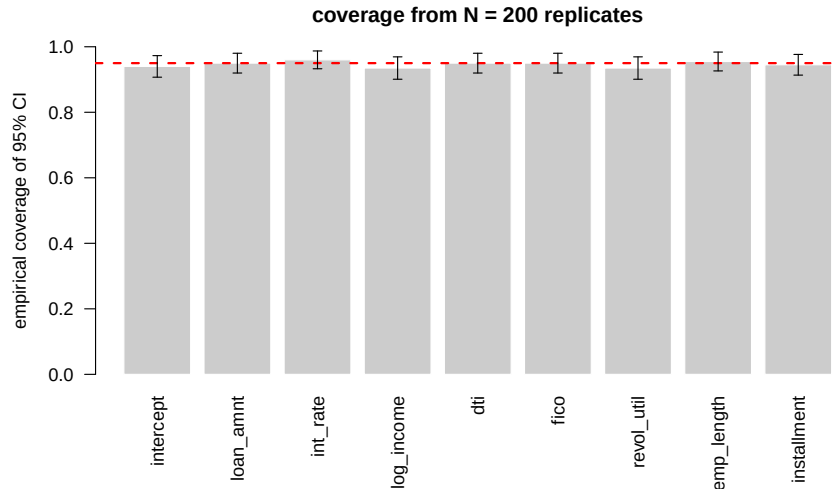


Figure 2: Empirical coverage of 95% credible intervals from $N = 200$ replicates. Dashed red line is the nominal 0.95 level. Error bars are ± 1.96 Monte Carlo standard errors.

Parameter	β^*	Mean of post. means	Bias	95% Coverage	MC SE
intercept	-1.575	-1.580	-0.005	0.940	0.017
loan_amnt	0.680	0.682	0.002	0.950	0.015
int_rate	0.518	0.522	0.004	0.960	0.014
log_income	-0.147	-0.148	-0.001	0.935	0.017
dti	0.149	0.150	0.001	0.950	0.015
fico	-0.262	-0.263	0.000	0.950	0.015
revol_util	-0.080	-0.082	-0.002	0.935	0.017
emp_length	-0.039	-0.040	-0.001	0.955	0.015
installment	-0.507	-0.508	-0.001	0.945	0.016

Table 2: Coverage and bias from $N = 200$ replicates. All coverage values fall within Monte Carlo error of the nominal 0.95.

The simulation results support the validity of the procedure. The procedure is approximately unbiased, the credible intervals have correct coverage, and the chains mix well. This provides a basis for trusting the posterior summaries from the real data fit.

7 Real Data Fit

For the real fit I drew a 10,000-row subsample from the cleaned data and ran the same MH sampler for 30,000 iterations with a 10,000-iteration burn-in. The sampler completed in approximately one

minute.

7.1 Diagnostics

Figure 3 shows trace plots for all nine parameters. The chains appear stationary throughout the post-burn-in window with no visible drift. Figure 4 shows posterior histograms, with the posterior mean (red) and 95% credible interval (blue dashed) marked.

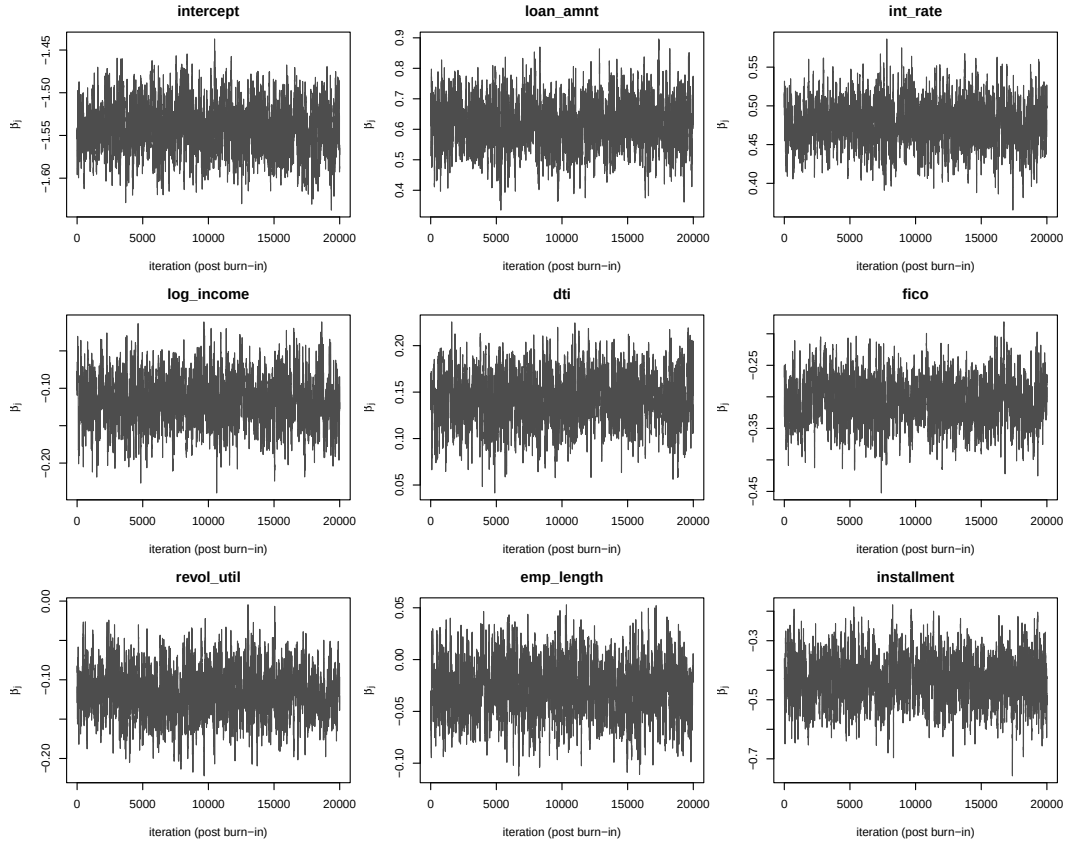


Figure 3: MH trace plots for the real data fit. Each panel shows 20,000 post-burn-in samples.

7.2 Interpretation

The signs of the coefficients are consistent with expectations for almost every covariate. Higher interest rate, higher debt-to-income, and larger loan amount are associated with higher default risk. Higher FICO, higher income, and higher revolving utilization are associated with lower default risk. Eight of nine credible intervals exclude zero. The exception is `emp_length`, with an interval of $[-0.076, 0.020]$, suggesting employment length adds little explanatory value once income and FICO

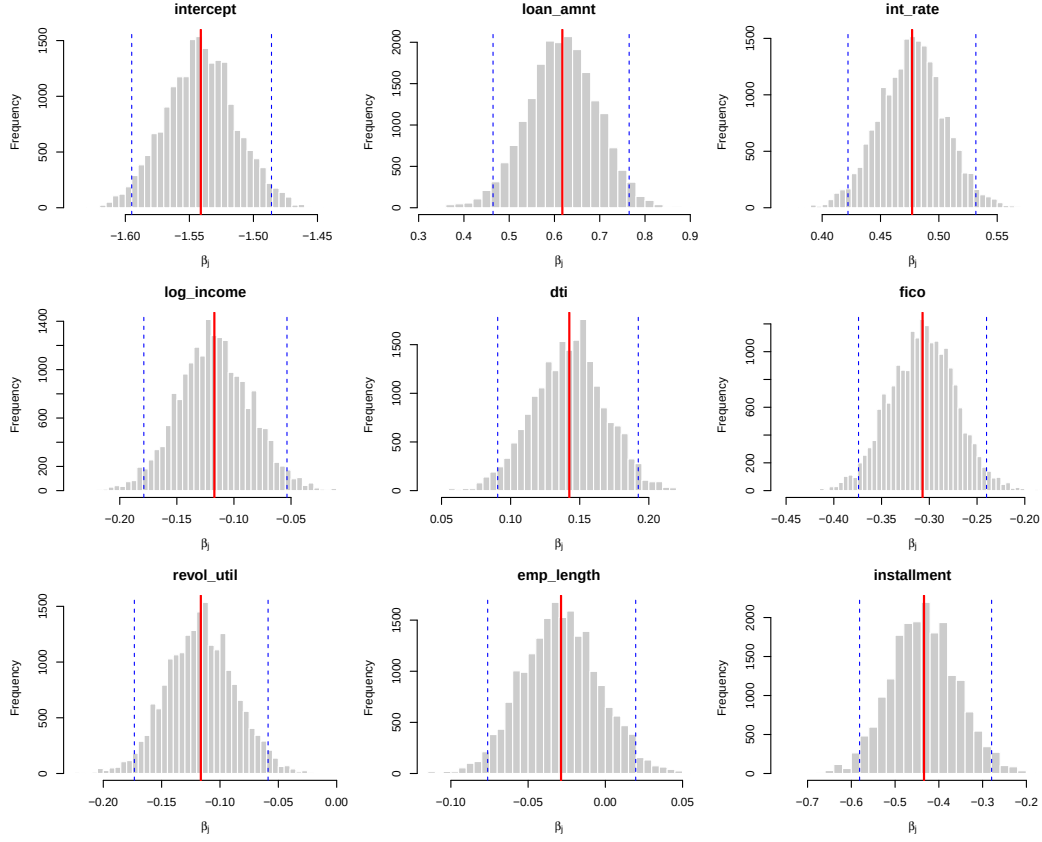


Figure 4: Posterior distributions on the real data. Red lines mark posterior means; blue dashed lines mark 95% credible interval bounds.

Parameter	Posterior mean	Median	2.5%	97.5%
intercept	-1.541	-1.541	-1.595	-1.486
loan_amnt	0.617	0.618	0.464	0.765
int_rate	0.477	0.478	0.422	0.532
log_income	-0.117	-0.118	-0.179	-0.054
dti	0.142	0.143	0.091	0.192
fico	-0.307	-0.307	-0.374	-0.240
revol_util	-0.116	-0.116	-0.173	-0.059
emp_length	-0.029	-0.029	-0.076	0.020
installment	-0.434	-0.434	-0.581	-0.279

Table 3: Posterior summaries from the real data fit.

are included.

The `installment` coefficient requires more discussion. The posterior mean is -0.434 , which would suggest that larger monthly payments are associated with lower default. This interpretation is not correct. The actual mechanism is that `installment` is a deterministic function of `loan_amnt`, `int_rate`, and the loan term (36 or 60 months). Once `loan_amnt` and `int_rate` are in the model, `installment` mostly carries information about term length: longer term means lower installment for the same loan amount, and 60-month loans default at higher rates than 36-month loans. The negative sign reads as “after holding amount and rate fixed, lower installments flag longer-term loans, which default more.” This is an issue of confounding, not an inference problem.

The simulation study supports confidence in the credible intervals reported here. Since the procedure was calibrated on synthetic data with the same dimensionality and scale, the intervals on real data should be interpreted at face value rather than treated with suspicion.

8 Limitations

The biggest issue is the omitted term variable. Loans come in 36-month and 60-month terms and term is one of the strongest predictors of default by itself. Adding it would probably flip the `installment` sign or shrink it substantially, which would be a more defensible result.

The model is linear in β on the log-odds scale and assumes additive effects. There is no strong reason to expect this to hold. A borrower with both a low FICO and a high DTI is probably worse than the sum of those two effects. Interactions or a nonlinear model would capture this.

The data only contains loans Lending Club agreed to fund, so the inferences apply to a population that already passed underwriting, not the full applicant pool. The selection process is not observable in the data.

Finally, the dataset spans 2007 through 2018. Loans issued in 2008 and 2017 were originated under very different macroeconomic conditions. The model has no time component, so the coefficients are an average over a long period that likely masks meaningful variation across vintages.

9 Conclusion

The Bayesian logistic regression recovers reasonable coefficients with credible intervals that the simulation study supports. The simulation study showed the posterior is calibrated under conditions matching the real fit, so the real-data intervals can be interpreted directly. The two changes most likely to improve the model are adding the loan term as a covariate and adding some treatment of vintage. Both are mechanical extensions, not changes to the inference framework.